

CHAPTER 15
Chemical Equilibrium

- Which of the following is true about a chemical system at equilibrium?
 - no reactions take place
 - temperature increases will no longer increase reaction rates
 - the rates of forward and reverse reactions are equal
 - all reaction products will be solids
- Which is true about the equilibrium constant expression?
 - it determines the activation energy needed to perform a reaction
 - it relates reactant and product concentrations
 - it relates concentrations to rates
 - it tells which reactant is highest in concentration
- Which of the following is the correct form of the equilibrium constant expression for the reaction:

$$2 \text{SO}_3(\text{g}) \rightleftharpoons 2 \text{SO}_2(\text{g}) + \text{O}_2(\text{g})$$
 - $K_c = [\text{SO}_2][\text{O}_2] / [\text{SO}_3]$
 - $K_c = [\text{SO}_3]^2 / [\text{SO}_2]^2[\text{O}_2]$
 - $K_c = [\text{SO}_2]^2[\text{O}_2] / [\text{SO}_3]^2$
 - $K_c = [\text{SO}_2][\text{O}_2]^2$
- Which of the following is the correct form of the equilibrium constant expression for the reaction below?

$$\text{NH}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{NH}_4^+(\text{aq}) + \text{OH}^-(\text{aq})$$
 - $K_c = [\text{NH}_4^+][\text{OH}^-] / [\text{NH}_3][\text{H}_2\text{O}]$
 - $K_c = [\text{NH}_3]$
 - $K_c = [\text{NH}_4^+][\text{OH}^-]$
 - $K_c = [\text{NH}_4^+][\text{OH}^-] / [\text{NH}_3]$
- Which of the following correctly relates the two equilibrium constants for the two reactions shown?

$$\text{NOCl}(\text{g}) \rightleftharpoons \text{NO}(\text{g}) + 1/2 \text{Cl}_2(\text{g}) \quad K_1$$

$$2 \text{NO}(\text{g}) + \text{Cl}_2(\text{g}) \rightleftharpoons 2 \text{NOCl}(\text{g}) \quad K_2$$
 - $K_2 = -K_1^2$
 - $K_2 = 1/(K_1)^{1/2}$
 - $K_2 = 1/K_1^2$
 - $K_2 = 2 K_1$
- The equilibrium for the following reaction at 700 K is, $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2 \text{HI}(\text{g})$ $K_1 = 55.17$
 What is the value of the equilibrium constant for the following reaction ($= K_2$)?

$$\text{HI}(\text{g}) \rightleftharpoons 1/2 \text{H}_2(\text{g}) + 1/2 \text{I}_2(\text{g}) \quad K_2 = ?$$
 - 3.29×10^{-5}
 - 55.17
 - 0.0181
 - 0.135
- Given the equilibrium constant, K_c , for the following reaction at 25°C, $\text{N}_2(\text{g}) + 3 \text{H}_2(\text{g}) \rightleftharpoons 2 \text{NH}_3(\text{g})$ is $K_c = 3.6 \times 10^8$. Calculate the equilibrium constant, K_p , for the same reaction at the same temperature.
 - 2.8×10^{-9}
 - 6.0×10^5
 - 2.2×10^{11}
 - 1.5×10^9
- Given the following two equilibria,

$$\text{NiCO}_3(\text{s}) \rightleftharpoons \text{Ni}^{2+}(\text{aq}) + \text{CO}_3^{2-}(\text{aq}) \quad K_1 = 6.6 \times 10^{-9}$$

$$\text{HCO}_3^-(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{CO}_3^{2-}(\text{aq}) + \text{H}_3\text{O}^+(\text{aq}) \quad K_2 = 4.8 \times 10^{-11}$$
 calculate the equilibrium constant for the following reaction.

$$\text{NiCO}_3(\text{s}) + \text{H}_3\text{O}^+(\text{aq}) \rightleftharpoons \text{Ni}^{2+}(\text{aq}) + \text{HCO}_3^-(\text{aq}) + \text{H}_2\text{O}(\text{l}) \quad K_3 = ?$$
 - 7.3×10^{-3}
 - 3.2×10^{-19}
 - 140
 - 1.8×10^{-9}
- We examine the following reaction at 250°C: $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$. At equilibrium we find $[\text{PCl}_5] = 3.4 \times 10^{-5} \text{ M}$, $[\text{PCl}_3] = 1.3 \times 10^{-2} \text{ M}$, and $[\text{Cl}_2] = 1.0 \times 10^{-4} \text{ M}$. Calculate the equilibrium constant, K_c , reaction.
 - 26
 - 5.1
 - 2.8×10^{-4}
 - 0.038
- We place an excess (more than can dissolve) of $\text{PbCl}_2(\text{s})$ into 100. mL of water and find that at equilibrium, $[\text{Cl}^-] = 0.032 \text{ M}$. Calculate K_c for the following reaction: $\text{PbCl}_2(\text{s}) \rightleftharpoons \text{Pb}^{2+}(\text{aq}) + 2 \text{Cl}^-(\text{aq})$
 - 5.1×10^{-4}
 - 3.9×10^{-5}
 - 4.8×10^{-3}
 - 1.6×10^{-5}

11. We place 1.00 g of $[\text{Ni}(\text{NH}_3)_6]\text{Cl}_2$ into 100. mL of water and find that at equilibrium, $[\text{NH}_3] = 0.0452 \text{ M}$. Calculate K_c for the following reaction: $[\text{Ni}(\text{NH}_3)_6]^{2+}(\text{aq}) \rightleftharpoons \text{Ni}^{2+}(\text{aq}) + 6 \text{NH}_3(\text{aq})$
 (a) 1.79×10^{-9} (b) 2.23×10^{-4} (c) 1.60×10^{-9} (d) 7.32×10^{-18}
12. We place 0.010 mol of $\text{N}_2\text{O}_4(\text{g})$ in a 2.0 L flask at 200°C . After reaching equilibrium, $[\text{N}_2\text{O}_4] = 0.0042 \text{ M}$. What is K_c for the following reaction? $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2 \text{NO}_2(\text{g})$
 (a) 1640 (b) 3.1×10^{-4} (c) 6.1×10^{-4} (d) 8.8×10^{-6}
13. We have a solution where $\text{Ag}(\text{CN})_2^-(\text{aq})$, $\text{CN}^-(\text{aq})$, and $\text{Ag}^+(\text{aq})$ are at equilibrium. If we know that $[\text{Ag}(\text{CN})_2^-] = 0.030 \text{ M}$ and that $[\text{CN}^-] = 0.10 \text{ M}$, what is $[\text{Ag}^+]$ in the solution?
 $\text{Ag}(\text{CN})_2^-(\text{aq}) \rightleftharpoons \text{Ag}^+(\text{aq}) + 2 \text{CN}^-(\text{aq}) \quad K_c = 1.8 \times 10^{-19}$
 (a) $5.4 \times 10^{-20} \text{ M}$ (b) $5.4 \times 10^{-10} \text{ M}$ (c) $5.4 \times 10^{-19} \text{ M}$ (d) $3.0 \times 10^{-12} \text{ M}$
14. Determine if the following system is at equilibrium, the reactant concentrations are too high, the product concentrations are too high, or if one simply cannot determine which with the information given:
 $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g}) \quad K_p = 11.5$
 $P(\text{PCl}_5) = 1.15 \text{ atm}$, $P(\text{PCl}_3) = 5.30 \text{ atm}$, $P(\text{Cl}_2) = 2.80 \text{ atm}$
 (a) at equilibrium (b) product concentrations too high
 (c) reactant concentrations too high (d) cannot be determined
15. We have the following equilibrium: $\text{PbCl}_2(\text{s}) \rightleftharpoons \text{Pb}^{2+}(\text{aq}) + 2 \text{Cl}^-(\text{aq})$. What will happen to $[\text{Pb}^{2+}]$ if some solid NaCl is added to the flask?
 (a) it will increase (b) it will decrease (c) it will not change (d) cannot tell
16. We have the following equilibrium: $\text{BaSO}_4(\text{s}) \rightleftharpoons \text{Ba}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq})$. What will happen to $[\text{Ba}^{2+}]$ if additional solid BaSO_4 is added to the flask?
 (a) it will increase (b) it will decrease
 (c) it will not change (d) cannot tell with the information provided
17. We have the following equilibrium: $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2 \text{NO}_2(\text{g}) \quad \Delta H_{\text{rxn}} = +150 \text{ kJ}$. What will happen to $[\text{N}_2\text{O}_4]$ if the temperature is increased?
 (a) it will increase (b) it will decrease
 (c) it will not change (d) cannot tell with the information provided
18. We have the following equilibrium: $2 \text{SO}_3(\text{g}) \rightleftharpoons 2 \text{SO}_2(\text{g}) + \text{O}_2(\text{g})$. What will happen to $[\text{SO}_3]$ if the temperature is increased?
 (a) it will increase (b) it will decrease (c) it will not change (d) cannot tell
19. We place 2.0 mol of PCl_5 in a 5.0 L flask and let the system come to equilibrium. What will the final concentration of Cl_2 be? $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g}) \quad K_c = 0.47$
 (a) 0.40 M (b) 0.26 M (c) 0.33 M (d) 0.023 M
20. We have the following equilibrium: $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$ where $K_c = 0.470$. We have a 1.00 L flask where $[\text{PCl}_5] = 0.150 \text{ M}$, $[\text{PCl}_3] = 0.280 \text{ M}$, and $[\text{Cl}_2] = 0.252 \text{ M}$. If we add 0.20 mol of $\text{Cl}_2(\text{g})$ to the flask, what will $[\text{PCl}_3]$ be when equilibrium is reached?
 (a) 0.231 M (b) 0.290 M (c) 0.329 M (d) 0.0486 M
21. H_2 gas, H_2S gas, and solid sulfur are placed in a flask and heated to 90°C . K at this temperature is 0.068. $\text{H}_2(\text{g}) + \text{S}(\text{s}) \rightleftharpoons \text{H}_2\text{S}(\text{g})$. You mix 17.5 moles of H_2 and 1.25 moles of H_2S in a 0.50-L flask at 25°C . Is the system at equilibrium? If not, in which direction does the reaction shift in order to establish equilibrium?
 (a) The concentrations given are equilibrium concentrations
 (b) The reaction is not at equilibrium, and so shifts left to achieve equilibrium.
 (c) The reaction is not at equilibrium, and so shifts right to achieve equilibrium.